

PUBLISHED: 15-AUG-2017
2015.0 RANGE ROVER (LG), 303-01

ENGINE - TDV8 4.4L DIESEL

LEFT CYLINDER HEAD (G1509167)

REMOVAL AND INSTALLATION

12.29.04	CYLINDER HEAD - ASSEMBLY - LEFT /ONE ONLY - RENEW	LEFT HAND DRIVE, 4400 CC, TDV8, WITH PARTICULATE FILTER	14.2	USED WITHINS	
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PART(S)

STEP	REPLACE PART / RENEW PART	PART NAME	
Installation Step 6	Renew Part	Cylinder head gasket	1
Installation Step 6	Renew Part	Cylinder head bolts	10
Installation Step 6	Renew Part	Cylinder head bolts	2
Installation Step 13	Renew Part	Fuel rail balance pipe	1

REMOVAL



NOTES:

- Some variation in the illustrations may occur, but the essential information is always correct.
- Removal steps in this procedure may contain installation details.
- Engine shown removed for clarity.

1.



WARNING:

Make sure to support the vehicle with axle stands.

Raise and support the vehicle.

2.

Disconnect the battery ground cable.

Refer to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).

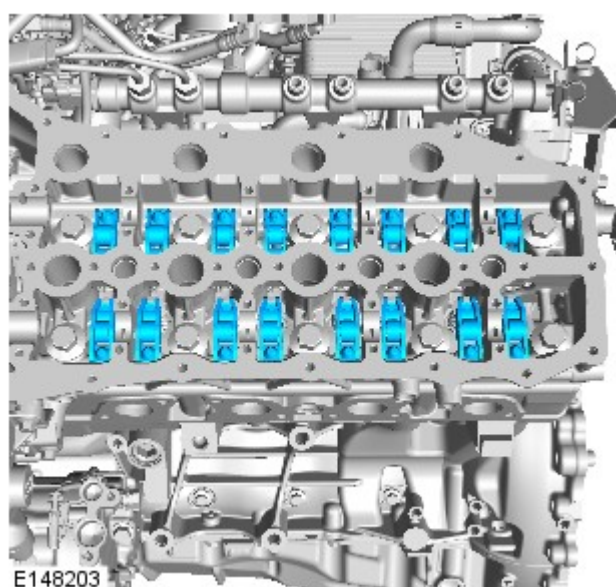
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3. Refer to: [Left Exhaust Manifold](#) (303-01E Engine - TDV8 4.4L Diesel, Removal and Installation).

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4. Refer to: [Generator](#) (414-02B Generator and Regulator - TDV8 4.4L Diesel, Removal and Installation).

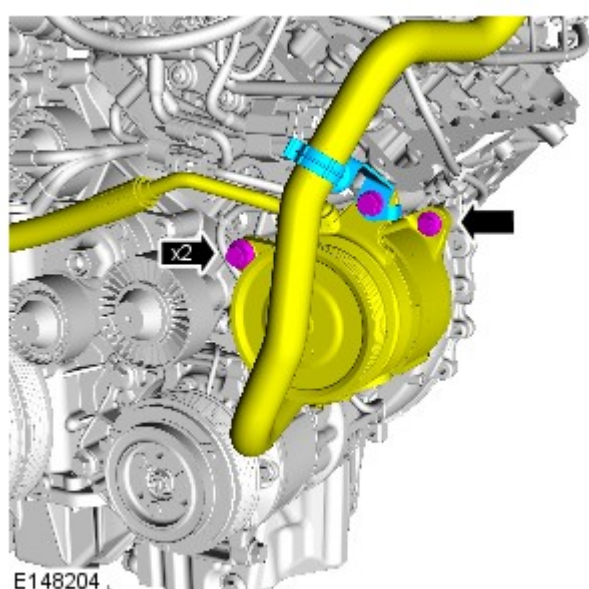
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5. Refer to: [Left Camshaft](#) (303-01E Engine - TDV8 4.4L Diesel, Removal and Installation).

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6.  **NOTE:**

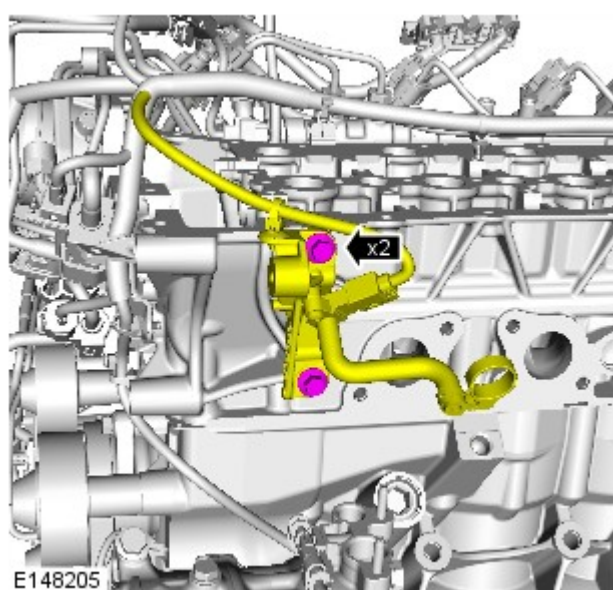
Note the position of the components prior to removal.



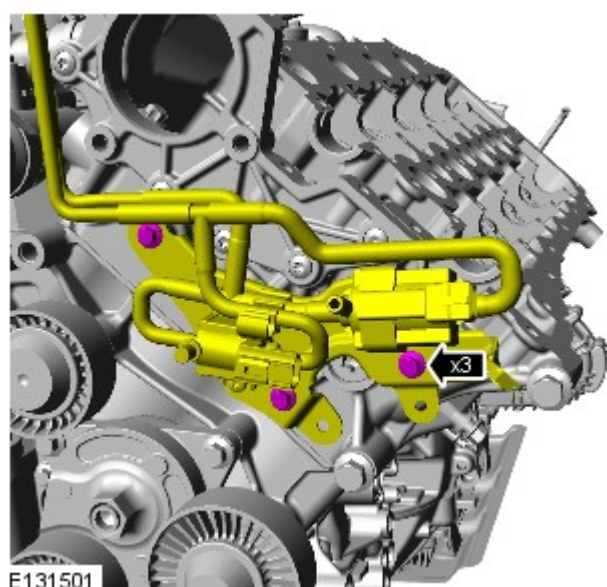
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- 7.



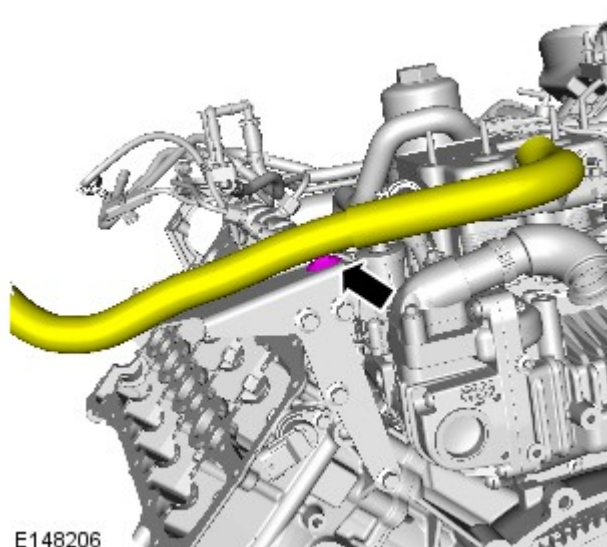
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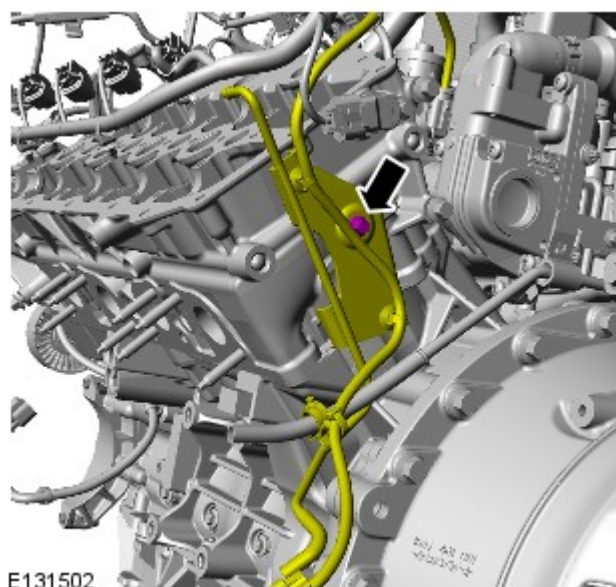
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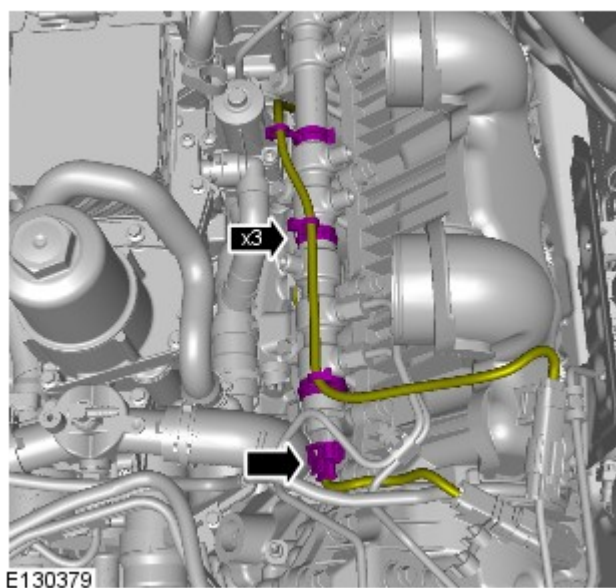
10.



11.



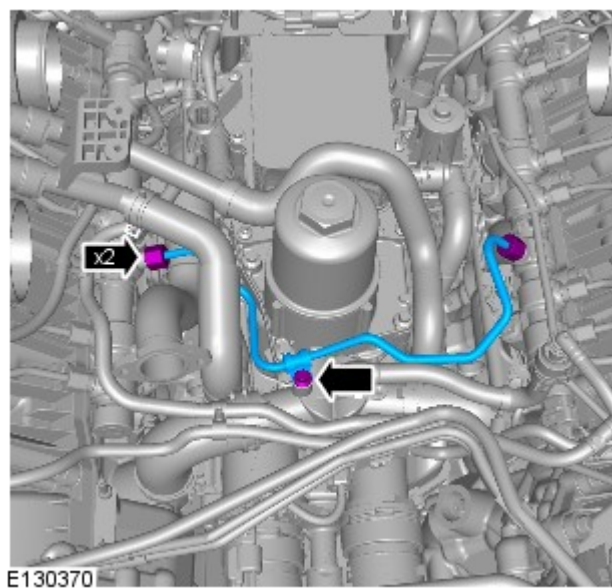
12.



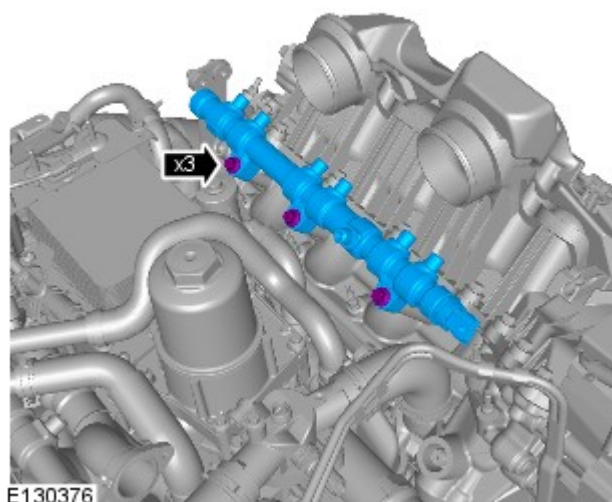
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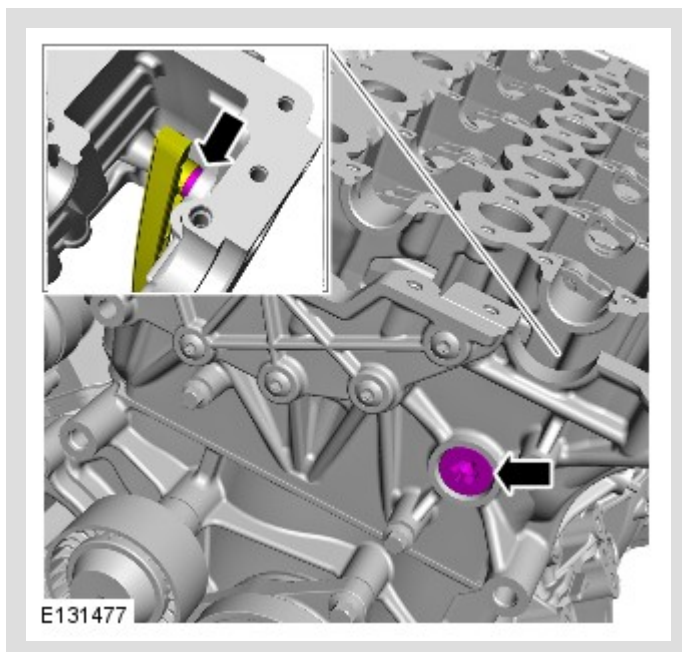
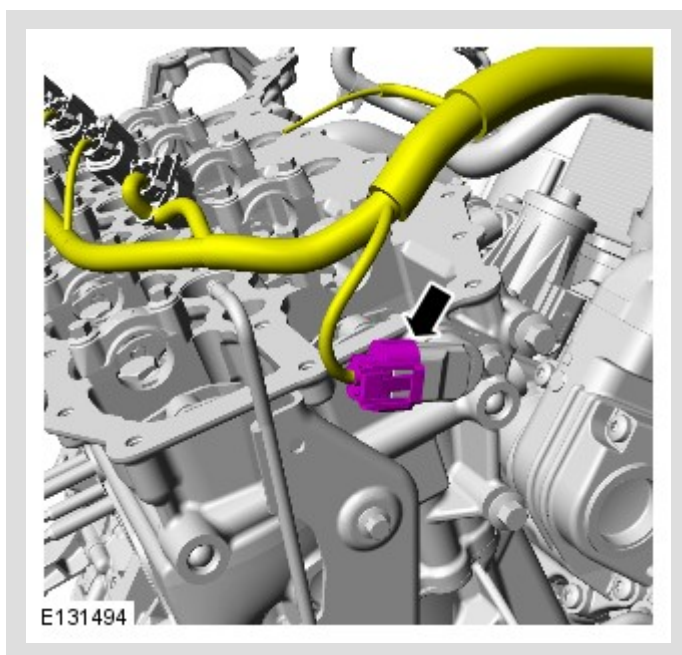
⚠ CAUTION:

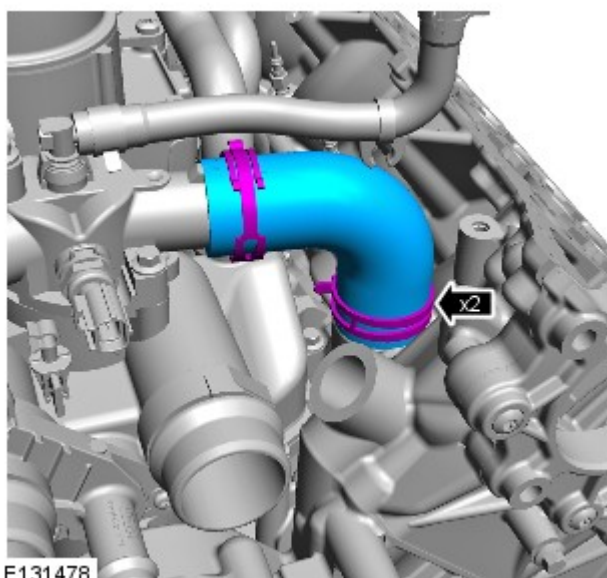
Discard the fuel pipes.



14.







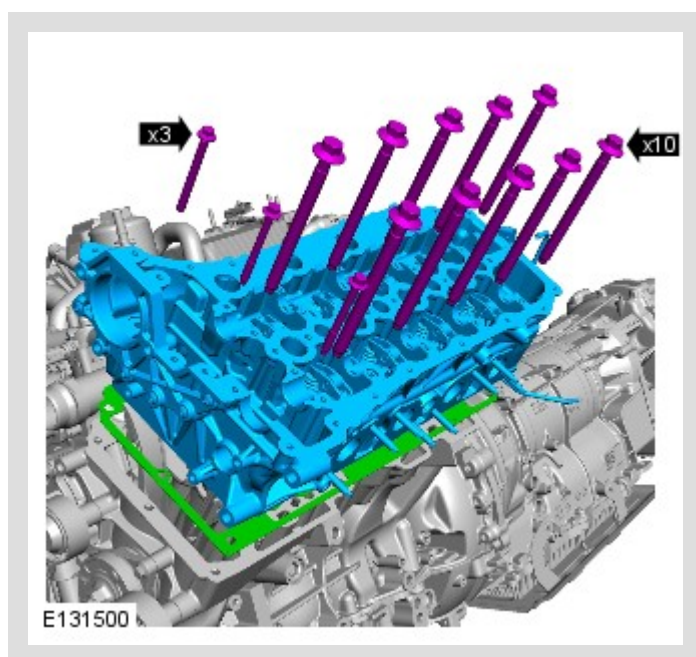
18.

⚠ CAUTION:

Discard the bolts.

△ NOTE:

Discard the gasket.

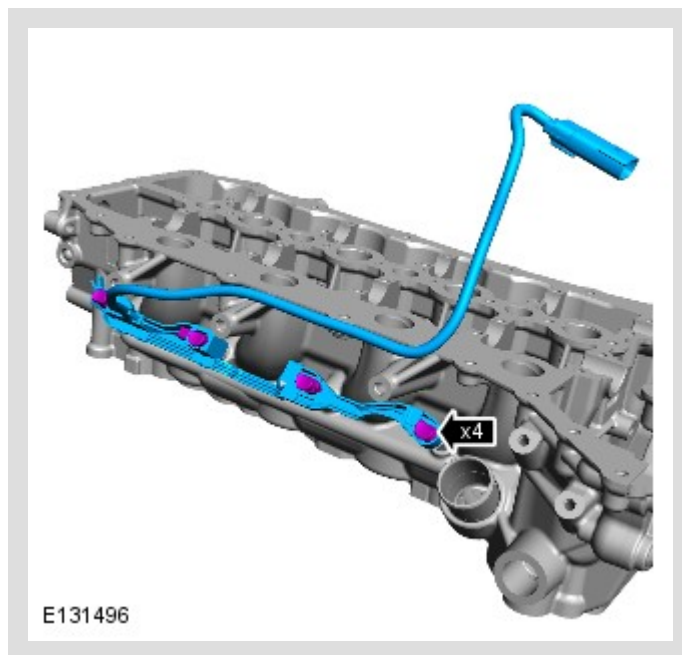


19.



CAUTION:

Take extra care not to damage the component.

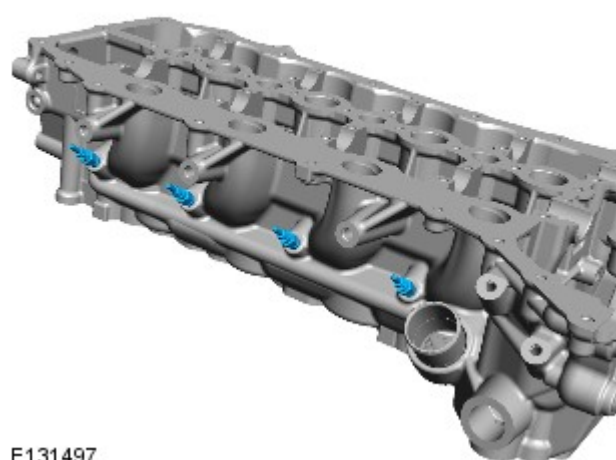


Do not disassemble further if removed for access only.

20.

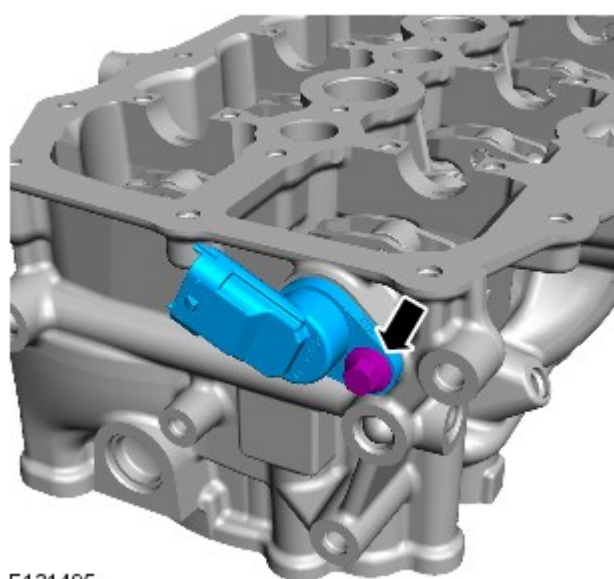
⚠ CAUTIONS:

- Take extra care not to damage the component.
- If accidentally dropped or knocked install a new glow plug.



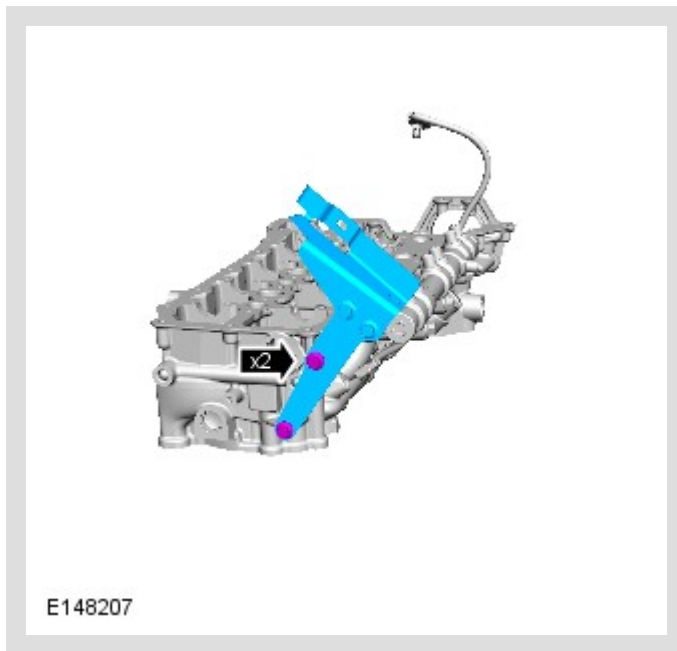
E131497

21.



E131495

22.

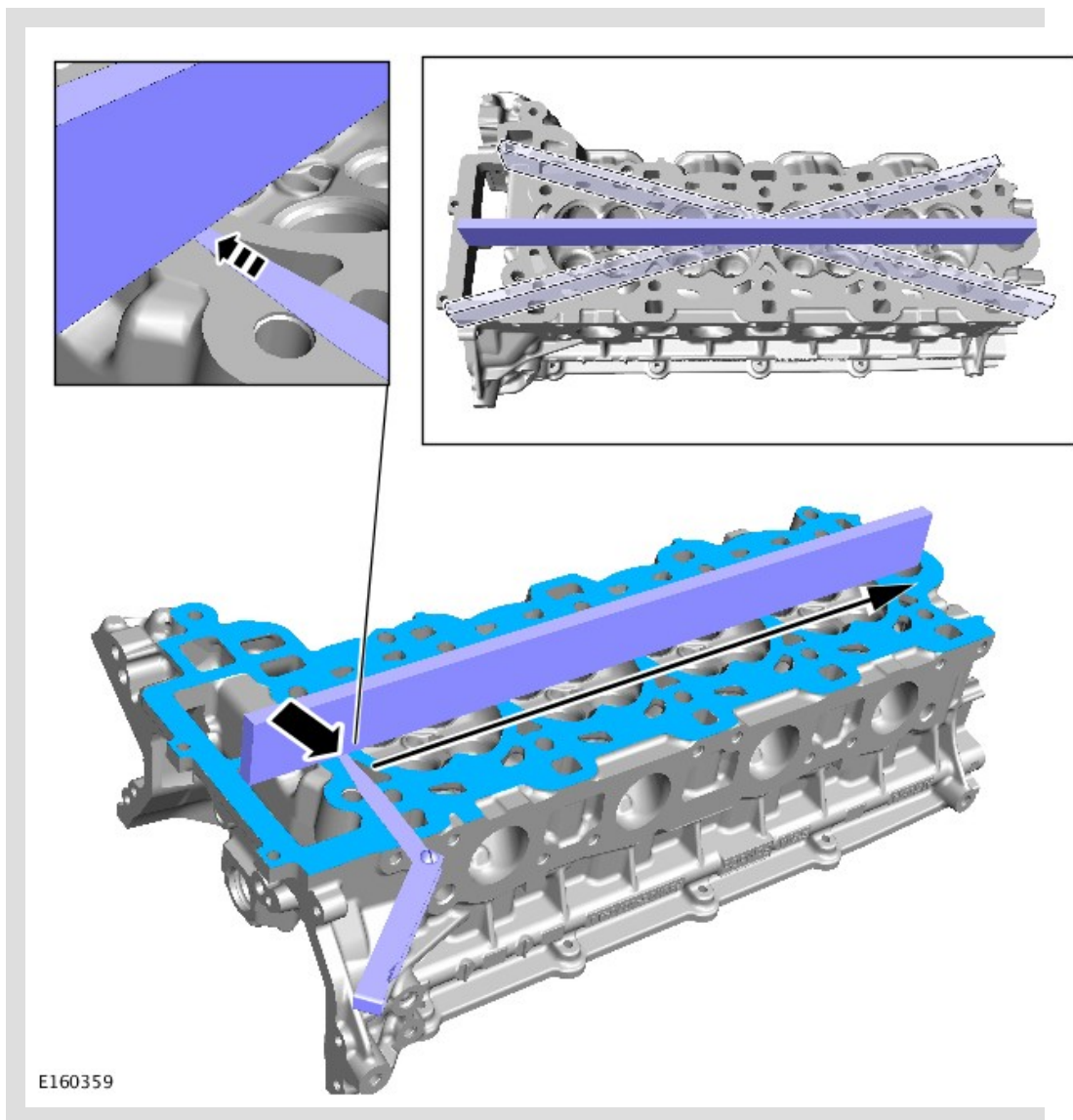


INSTALLATION

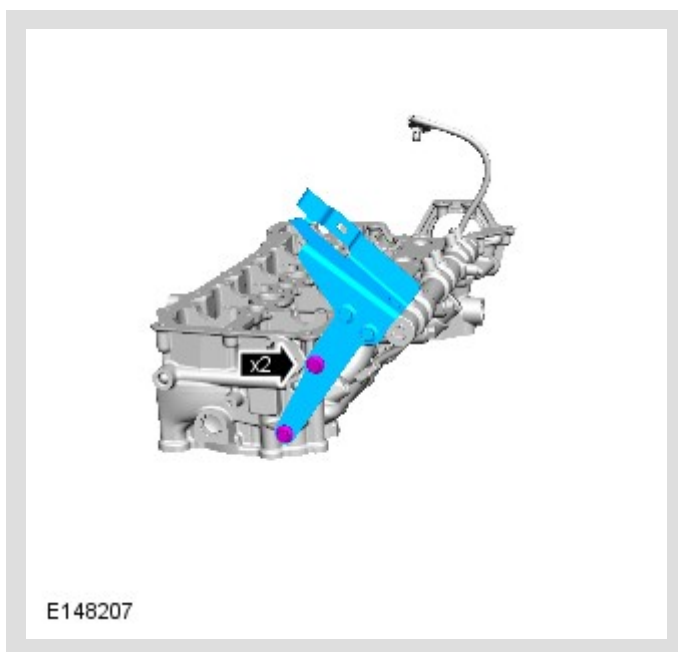
1.

⚠ CAUTIONS:

- Make sure that the mating faces are clean and free of foreign material.
- An acceptable flatness of the cylinder head is 0.2mm.

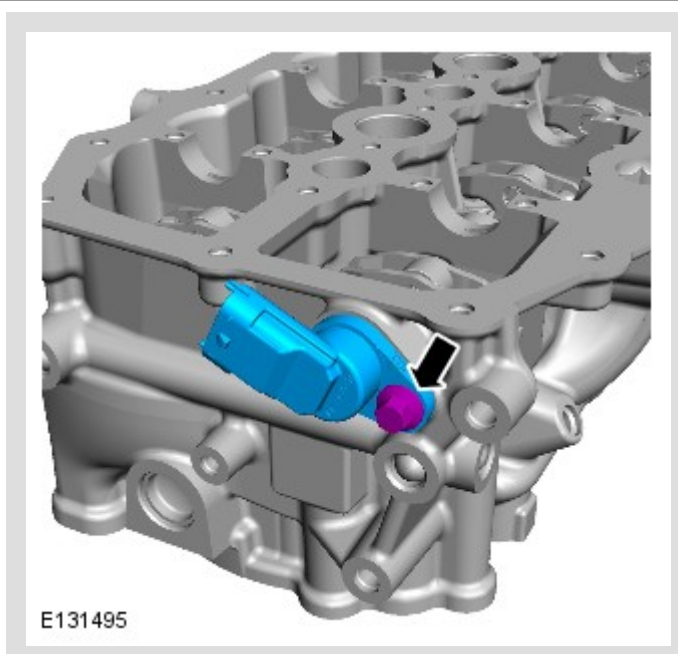


- Clean and inspect the cylinder head and cylinder block.
 - Check the cylinder head face for distortion, across the center and from corner to corner.
- Refer to: [Cylinder Head Distortion](#) (303-00 Engine System - General Information, General Procedures).



Torque: 25 Nm

3.



Torque: 13 Nm

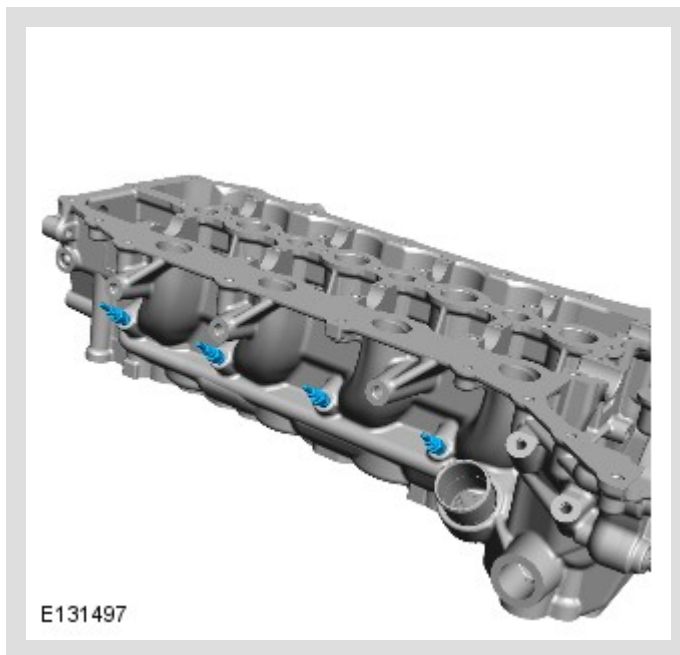
4.



CAUTIONS:

- Take extra care not to damage the component.

- If accidentally dropped or knocked install a new glow plug.

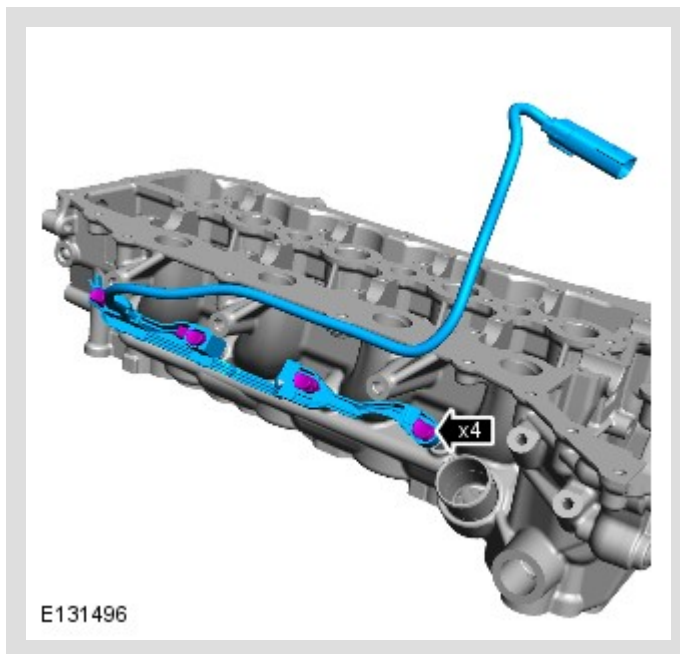


Torque: 10 Nm

5.

⚠ CAUTION:

Take extra care not to damage the component.



6.



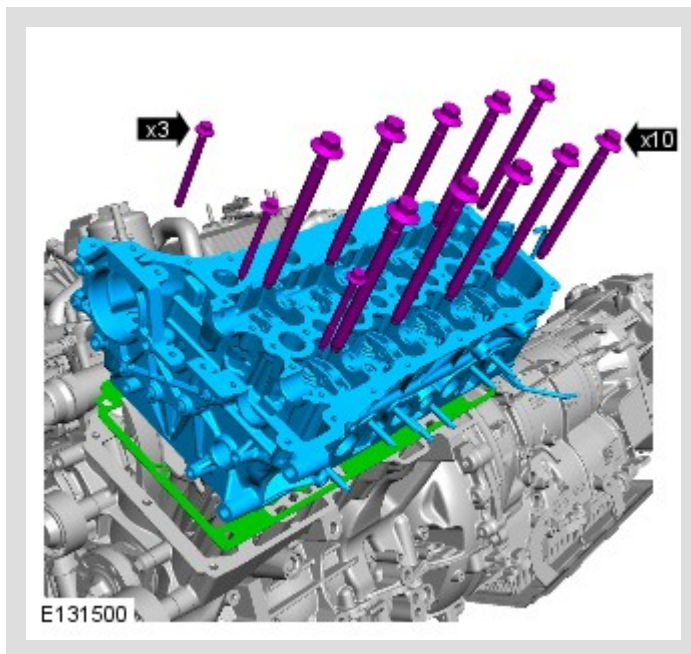
CAUTIONS:

- Make sure that the mating faces are clean and free of foreign material.
- Make sure that new cylinder head bolts are installed.
- Only tighten the bolts finger-tight at this stage.



NOTE:

Install a new cylinder head gasket.



Renew Part: *Cylinder head gasket* : 1 .

Renew Part: *Cylinder head bolts* : 10 .

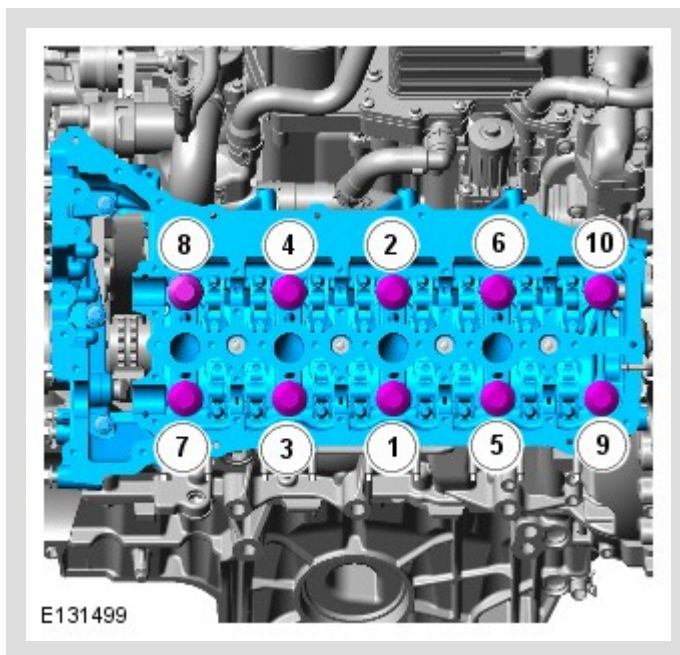
Renew Part: *Cylinder head bolts* : 2 .

7.



CAUTION:

Tighten the bolts in the sequence shown.



Torque:

Stage 1: 20 Nm

Stage 2: 40 Nm

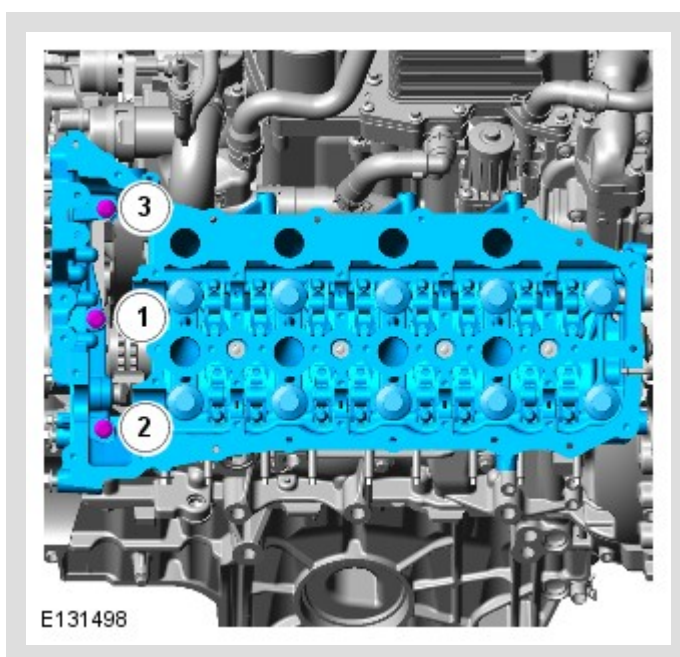
Stage 3: 80 Nm

Stage 4: 180°

8.

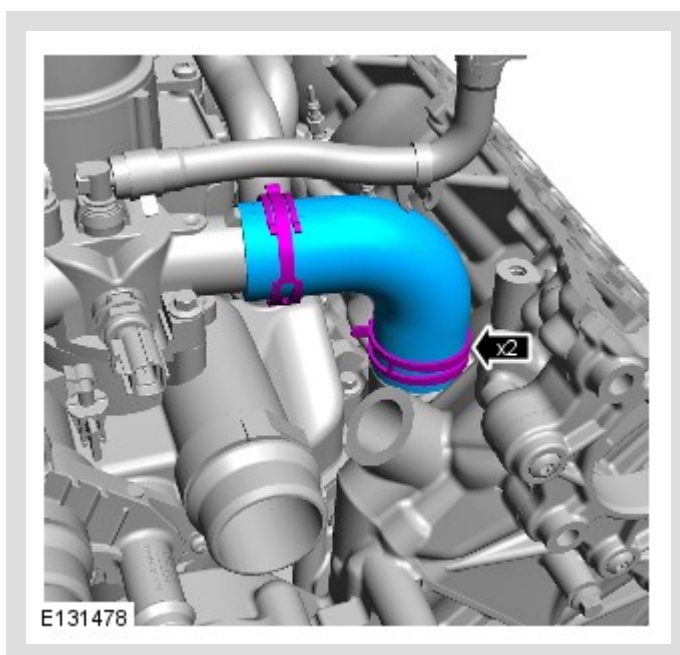
⚠ CAUTION:

Tighten the bolts in the sequence shown.

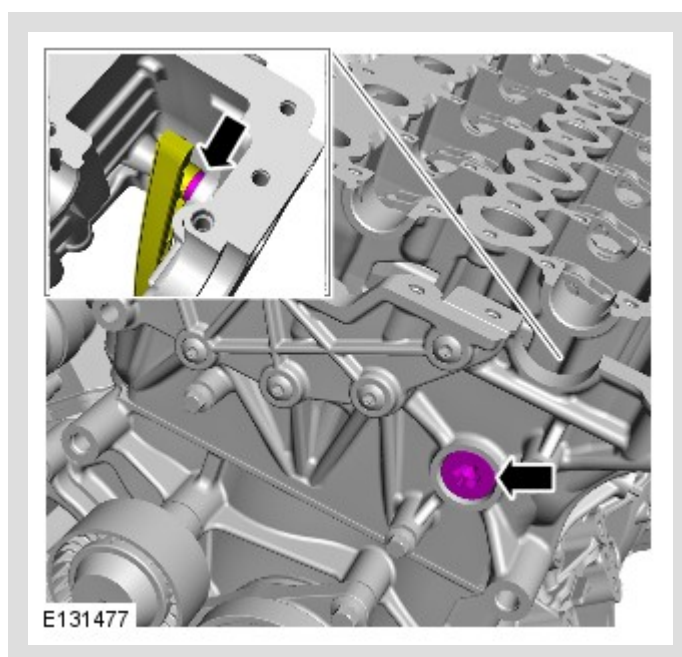


Torque: 20 Nm

9.

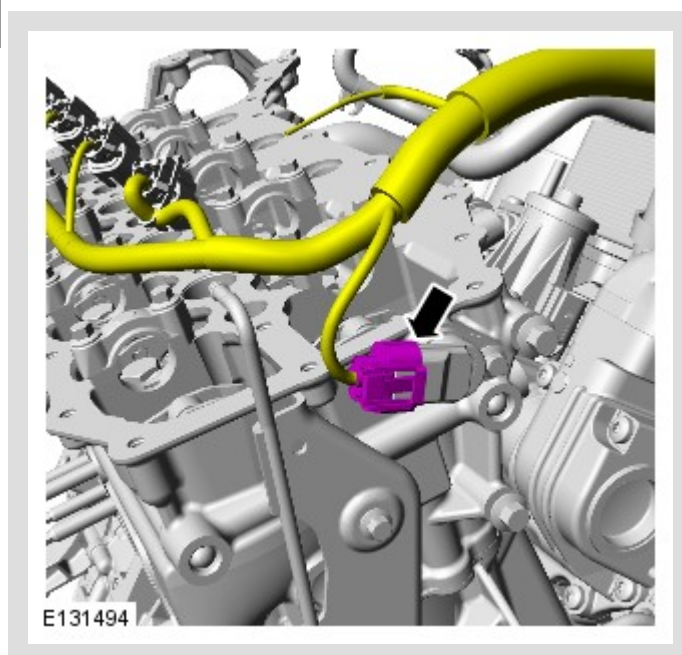


10.

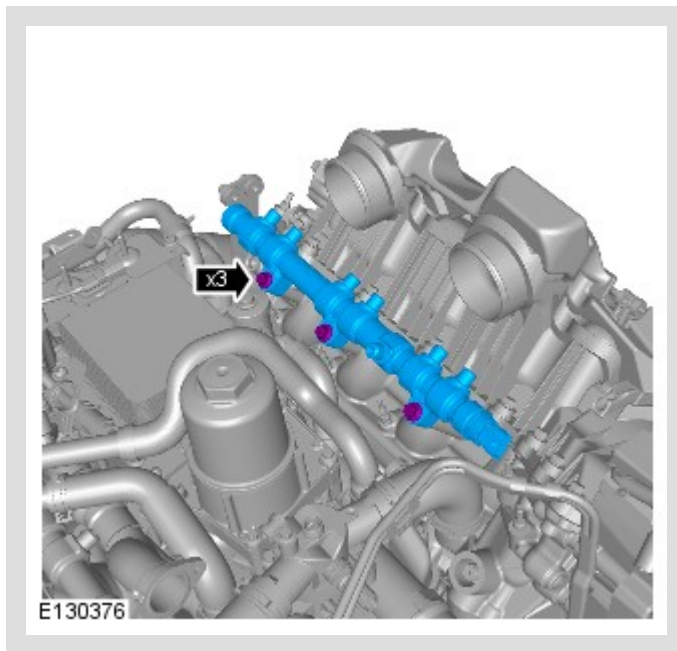


Torque: 27 Nm

11.



12.



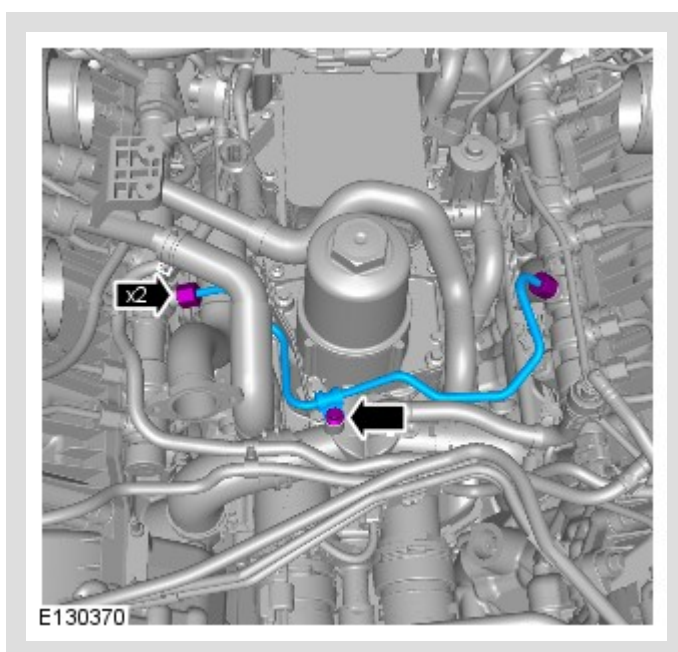
Torque: 23 Nm

13.



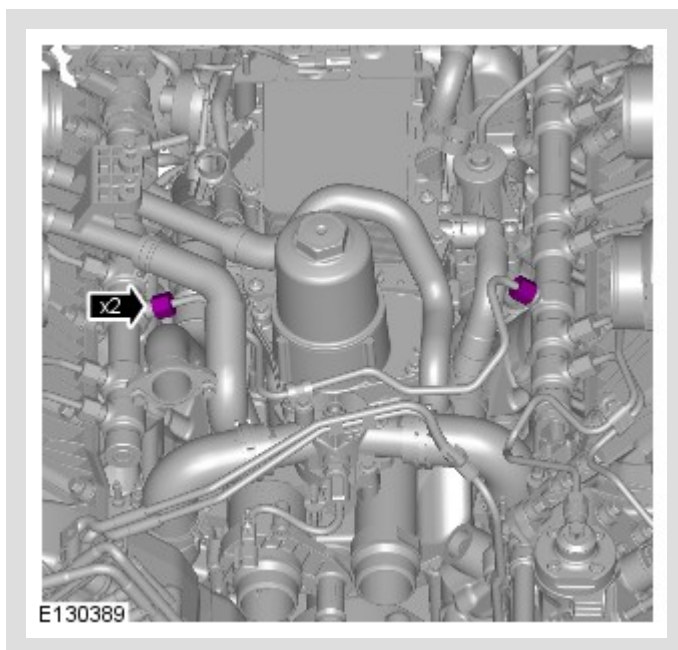
CAUTIONS:

- Install a new fuel pipe.
- Only tighten the unions finger-tight at this stage.
- Only tighten the bolts finger-tight at this stage.

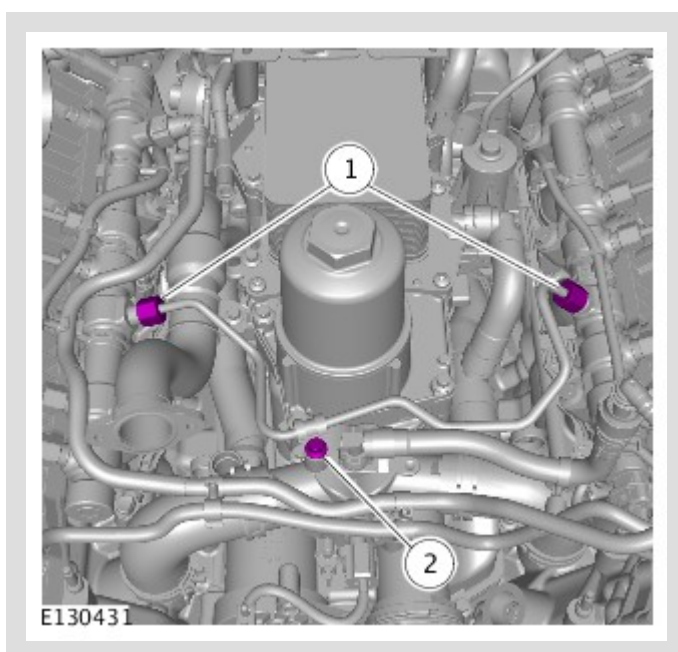


Renew Part: *Fuel rail balance pipe* : 1 .

14.



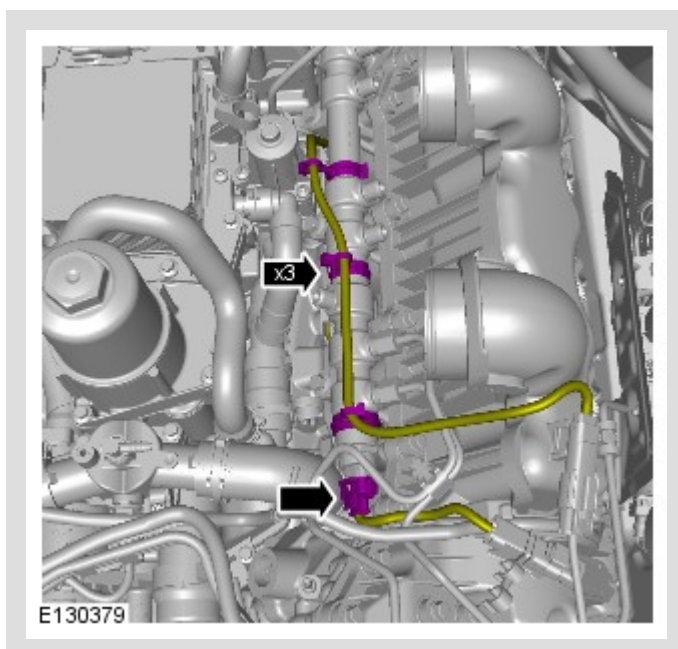
Torque: 10 Nm



Torque:

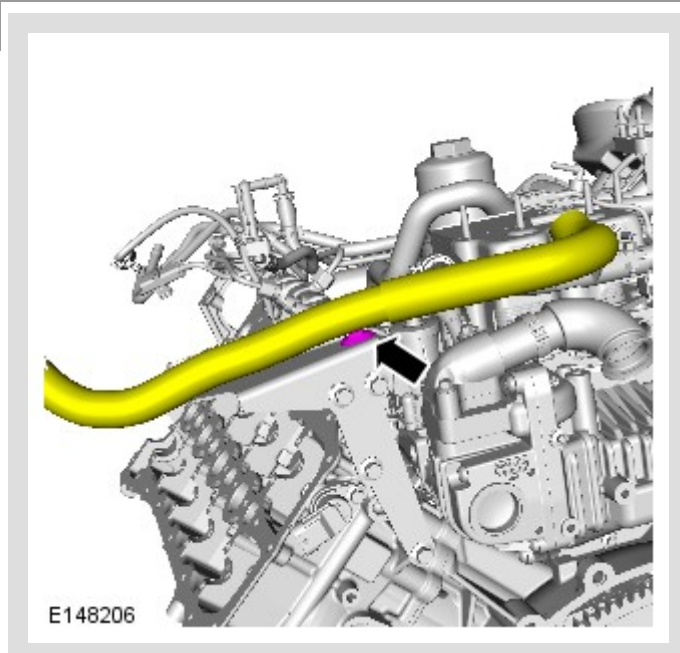
Unions 73°

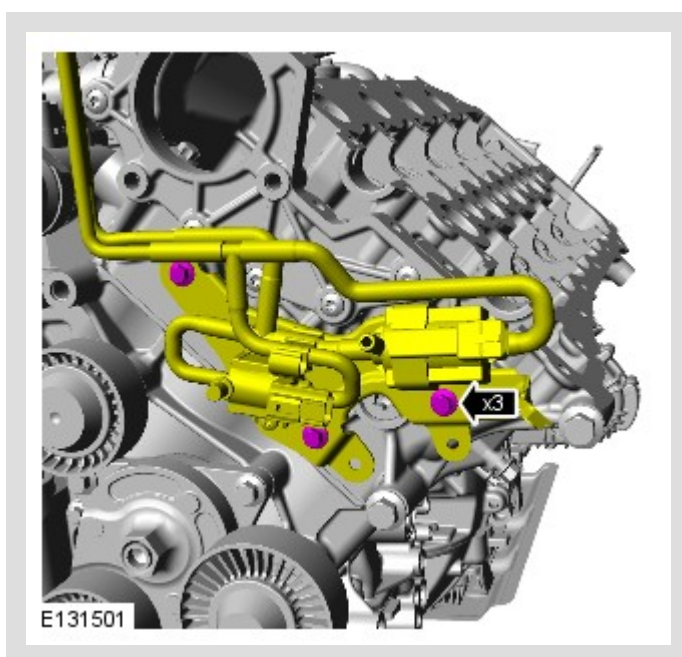
M6 10 Nm



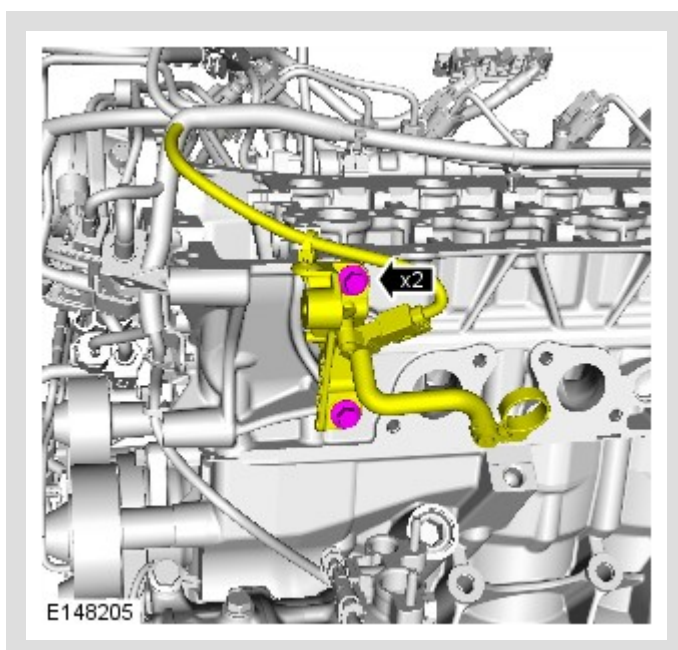


Torque: 10 Nm

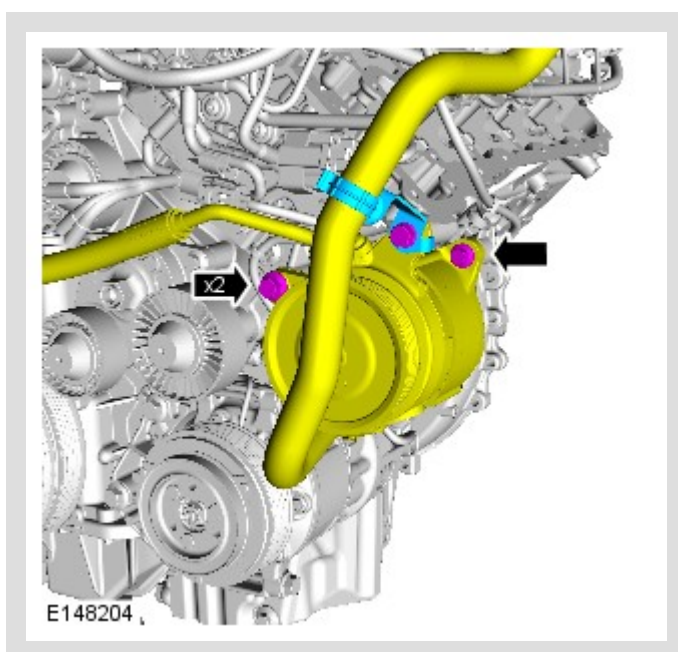




Torque: 10 Nm



Torque: 25 Nm



Torque:

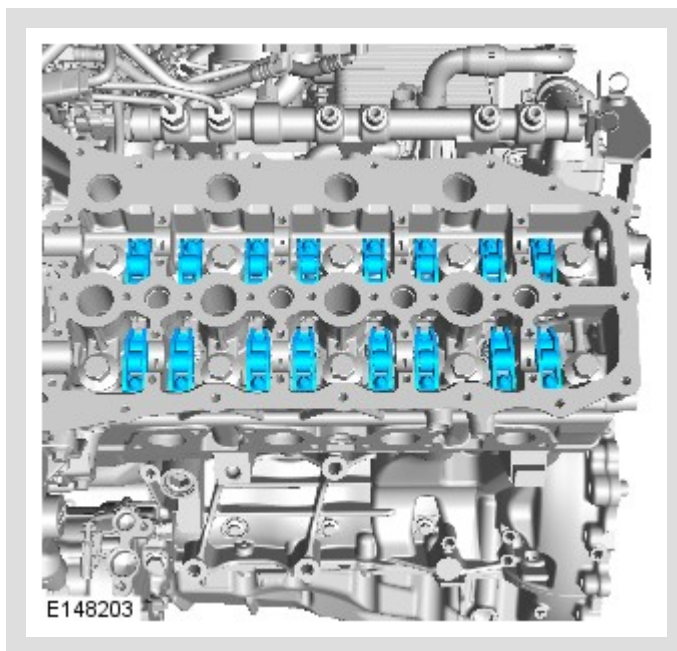
M8 25 Nm

M10 45 Nm



CAUTION:

Make sure that the components are installed to the noted position in the removal steps.



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23. Refer to: [Left Camshaft](#) (303-01E Engine - TDV8 4.4L Diesel, Removal and Installation).
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24. Refer to: [Generator](#) (414-02B Generator and Regulator - TDV8 4.4L Diesel, Removal and Installation).
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25. Refer to: [Left Exhaust Manifold](#) (303-01E Engine - TDV8 4.4L Diesel, Removal and Installation).
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26. Connect the battery ground cable.
Refer to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).

